



Dark matter gamma rays

Two new studies suggest that Dark matter is the source of the excessive Gamma Rays at galactic centers. <http://digit.in/apr21-61>



Mission pandora

NASA's concept mission Pandora could be the key to unlock the atmospheric mysteries of different worlds. <http://digit.in/apr21-62>

SAFER TELECOMMUNICATIONS

Mobile phones are some of the largest contributors to the sea of electromagnetic radiation, and the nature of cellular systems requires many base stations to give coverage to the areas served. The need to offer good coverage has to be offset against the need to limit the public from prolonged exposure to stray radiation. The effective management of this electromagnetic pollution requires tools to optimise the radio-frequency levels generated by these base stations. Space Engineering SpA., an Italian company, has been developing ways to analyse and model electromagnetic fields from spacecraft antennae to study their effects on nearby equipment – since 1989. This is used to check antennae performance which are vital to avoid malfunctions in on-board electronic systems. In 1993, they successfully turned their expertise in satellites to modelling complicated conditions experienced by mobile receivers in urban locations. These models were validated at the ESTEC Compact Antenna Test Range, which led Space Engineering to spin-off Teleinformatica e Sistemi (TeS) in 1998. This new company aimed to use its know-how from space work in the rapidly expanding telecommunications market. Their eventual system, Quickplan, has been developed to assist the needs of radio system developers, as well as environmental agencies. Quickplan is capable of calculating and displaying radio-frequency field levels across highly complex urban environments, indicating optimum locations of transmitters and the resulting electromagnetic pollution. It does this by drawing upon multiple maps to create a 3D image of the territory. It features editing and zoom capabilities which result in a recognisable 3D map of the location with colour coding to identify regions of high radio-frequency power levels. The success of this technology has enabled planners of radio systems to minimise their environmental impact and improve the servicing of telecommunication network coverage.

CAMERAS FOR OIL

Oil is one the world's most important natural resources on our planet. They provide energy and oil companies are always looking for ways to cut costs and improve efficiency. One such potential aid is being developed by Proneta – a UK based company. This is in the form of a camera that is capable of viewing things through oil, which would make remote inspections in oil wells and pipelines a lot easier – it can now be done without having to clean out the oil first. Normally, you would use a conventional video camera to analyse and diagnose problems in drilling and other installations. However, these cameras are unable to see through the oil, and hence can only be used once all of the oil has been flushed out with a continuous source of brine. The cost of flushing is massive, and the time delay caused due to transporting the pump and filter equipment to the rig could potentially mean that bringing a “down-hole” camera is just too expensive. Proneta's solution to this was to use its vast experience in the space industry, and in designing electro-optic instruments, to design and develop a camera with sensors capable of penetrating through oil. To demonstrate the capabilities of this camera, Proneta had to work beyond just paper calculations and computer modeling. With part-sponsorship from ESA, Proneta built a test-rig in their labs to produce pictures to market their product to all the major oil companies and other prospective sponsors. Their key patent for the new technology has been fully granted, and four major oil companies – Shell, BP, Amerada Hess and Chevron – fully support the development of a full-scale demonstrator, in which the cameras will be shown working with real-life situations. Proneta has come far in its journey, and has expanded its premises to accommodate its growing team and development facilities.

RADIATION DETECTING CHIPS

Space is a place filled with dangerous radiation, and spacecrafts carry instruments to measure and record

the amount of radiation around them. This is primarily to record the damage it does to electronics. One such device is called RADFET (radiation-sensitive field-effect transistor) which is a silicon chip developed in the late 1970s by a UK company – REM (Oxford) Ltd. This device monitors the cumulative radiation and has been incorporated in spacecraft such as Meteosat-3, as well as the Hubble Space Telescope. The chip records the amount of radiation as well as converts it into a radio signal, which makes it possible to communicate the radiation dose to mission controllers. This level of remote reading offers a huge avenue for potential applications on Earth, where remote sensing is necessary in areas such as nuclear safety and medicine. One such application is to use the chip as a dosimeter for monitoring radiotherapy used in treating cancer and heart disease. This is possible because the sensor is small, and is cheap to make. Previous radiation monitors were often expensive and unable to record data in real-time. Harvard Medical School used RADFET sensors and inserted them inside tumours in mice, before they received radiation treatment. The results encouraged REM to pursue further medical applications, and by partnering with the Istituto Scientifico Tumori (IST) the company explored a new use for radiation dosimeters in treating blocked arteries. Sometimes, an artery can be unblocked by inflating a balloon catheter to expand the blood vessels. Unfortunately, blockage can re-occur but recent research shows that this can be prevented by irradiation, ideally with beta-rays. This process is called endovascular brachytherapy, and involves inserting a radioactive source into the blood vessel via a catheter. One problem is that the radiation needs to be monitored carefully to prevent damage to the surrounding tissue. The RADFET offers a good solution, and is cheap enough to be thrown away after use. This is why this small space-derived device could develop into an important tool in the field of medicine. 